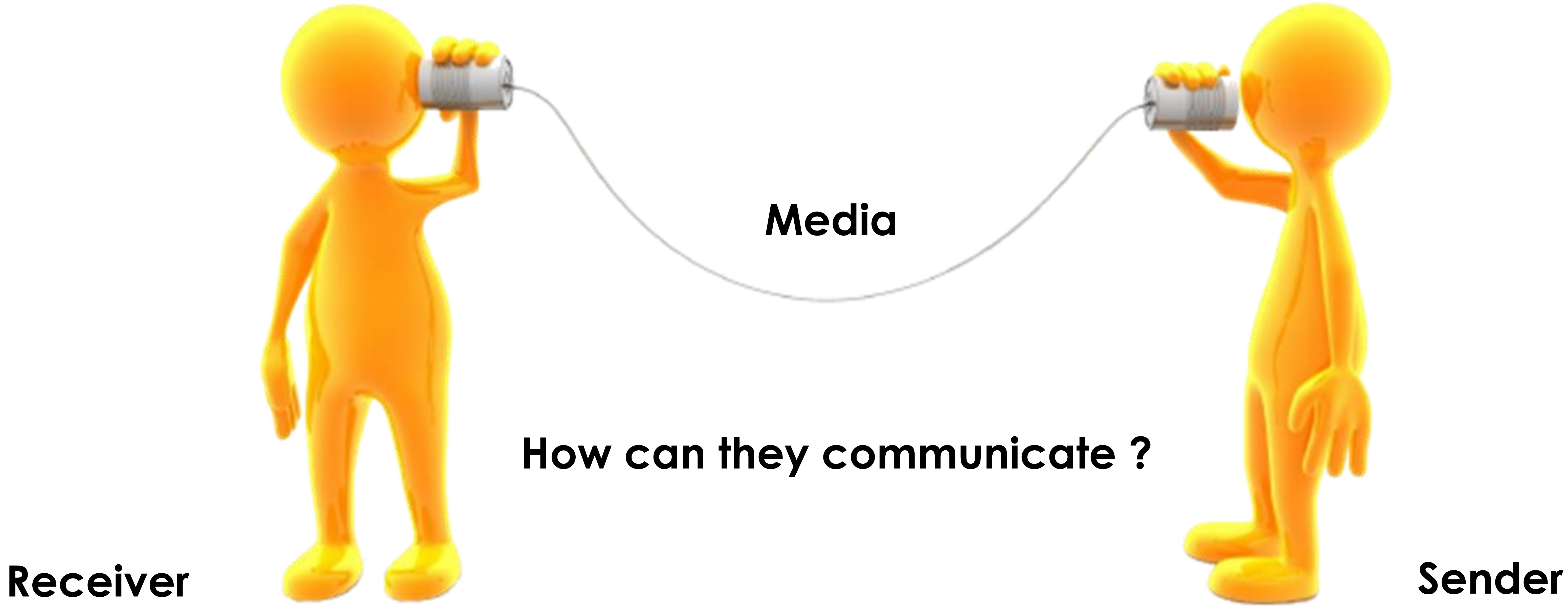
INSTITUT ASIA

CHAPTER 2

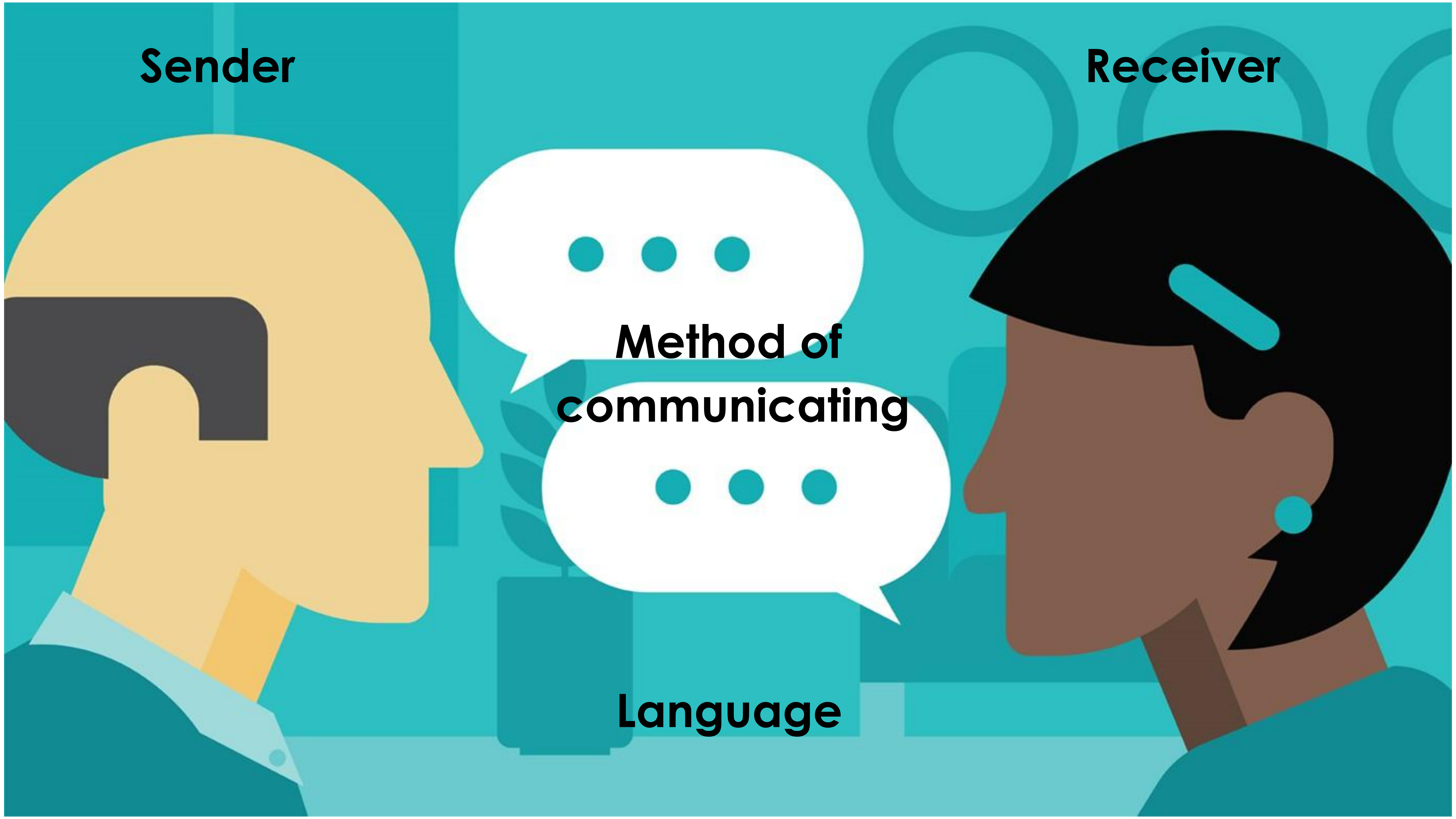
NETWORK COMMUNICATION & PROTOCOLS

JARINGAN KOMPUTER (MUT-14)
Hadi M.Kom

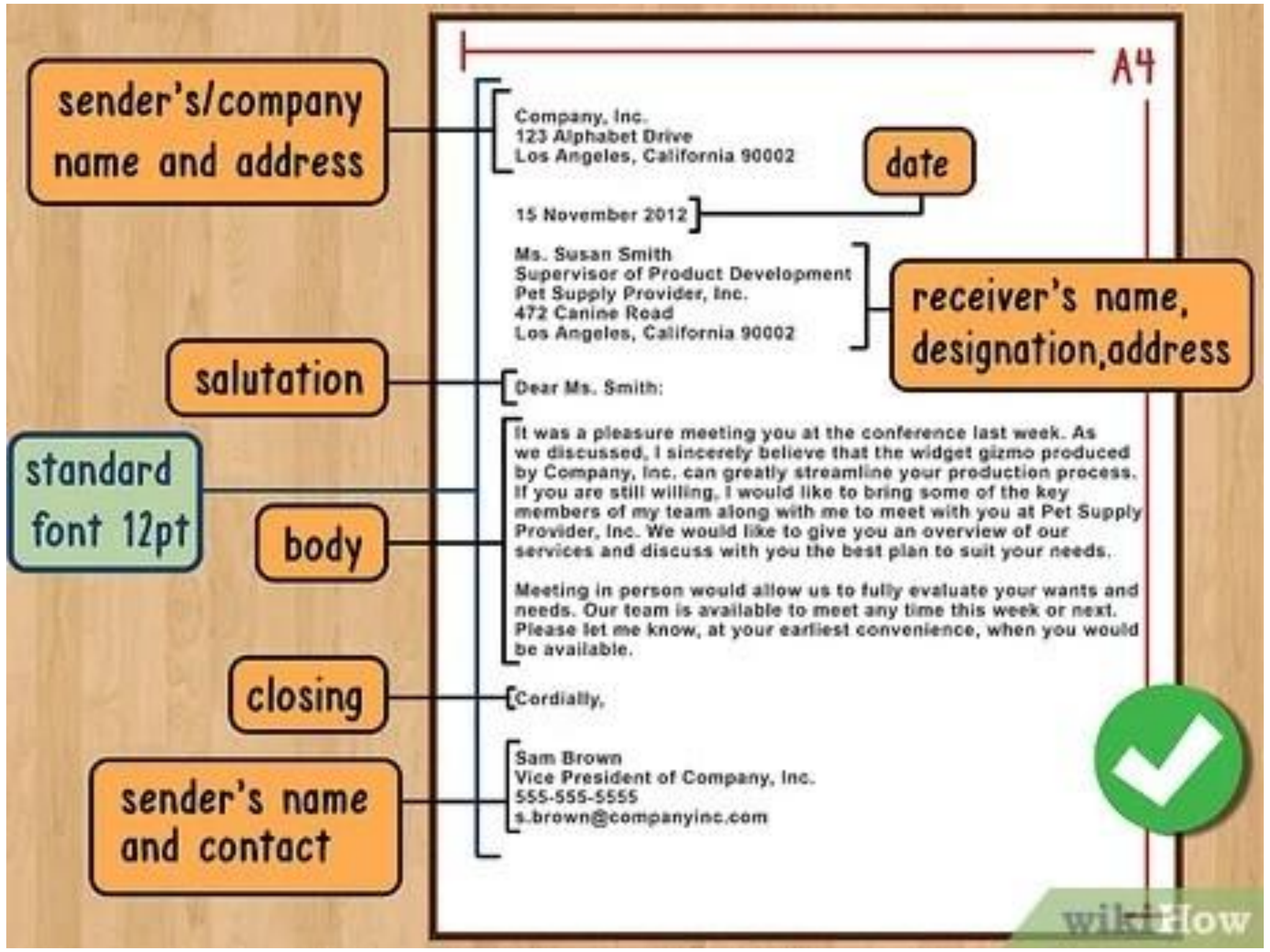
Communication Analogy



Rules of Communication



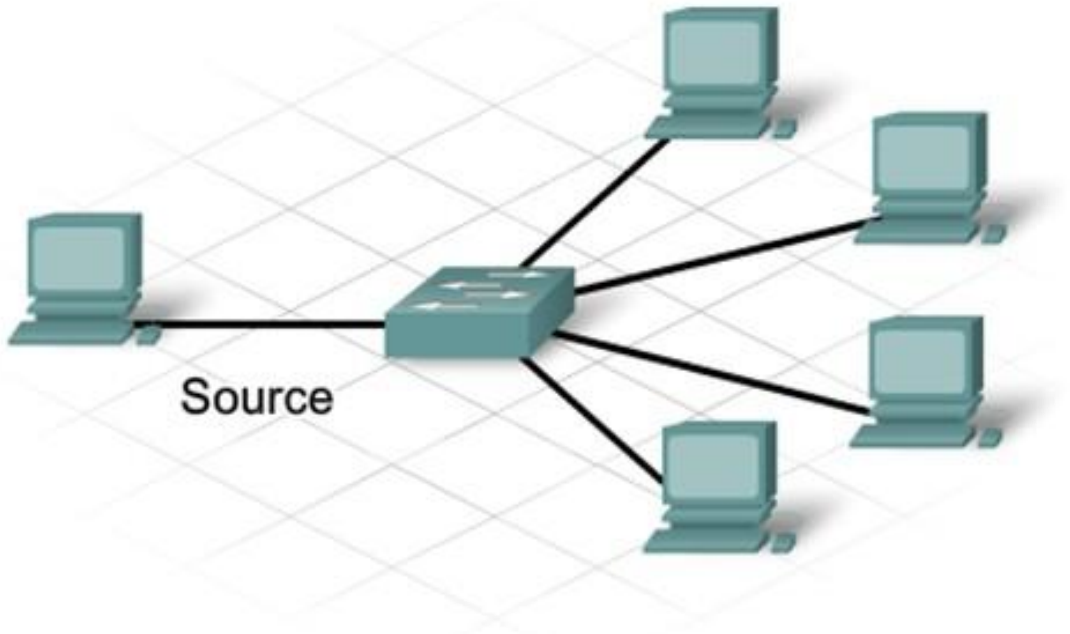
Rules of Communication



Part of Letter (DATA)

1. Sender's identity -- sender address
2. Receiver's identity -- receiver address
3. Salutation -- data header
4. Message content -- main data
5. Closing -- data header

Message Delivery Options

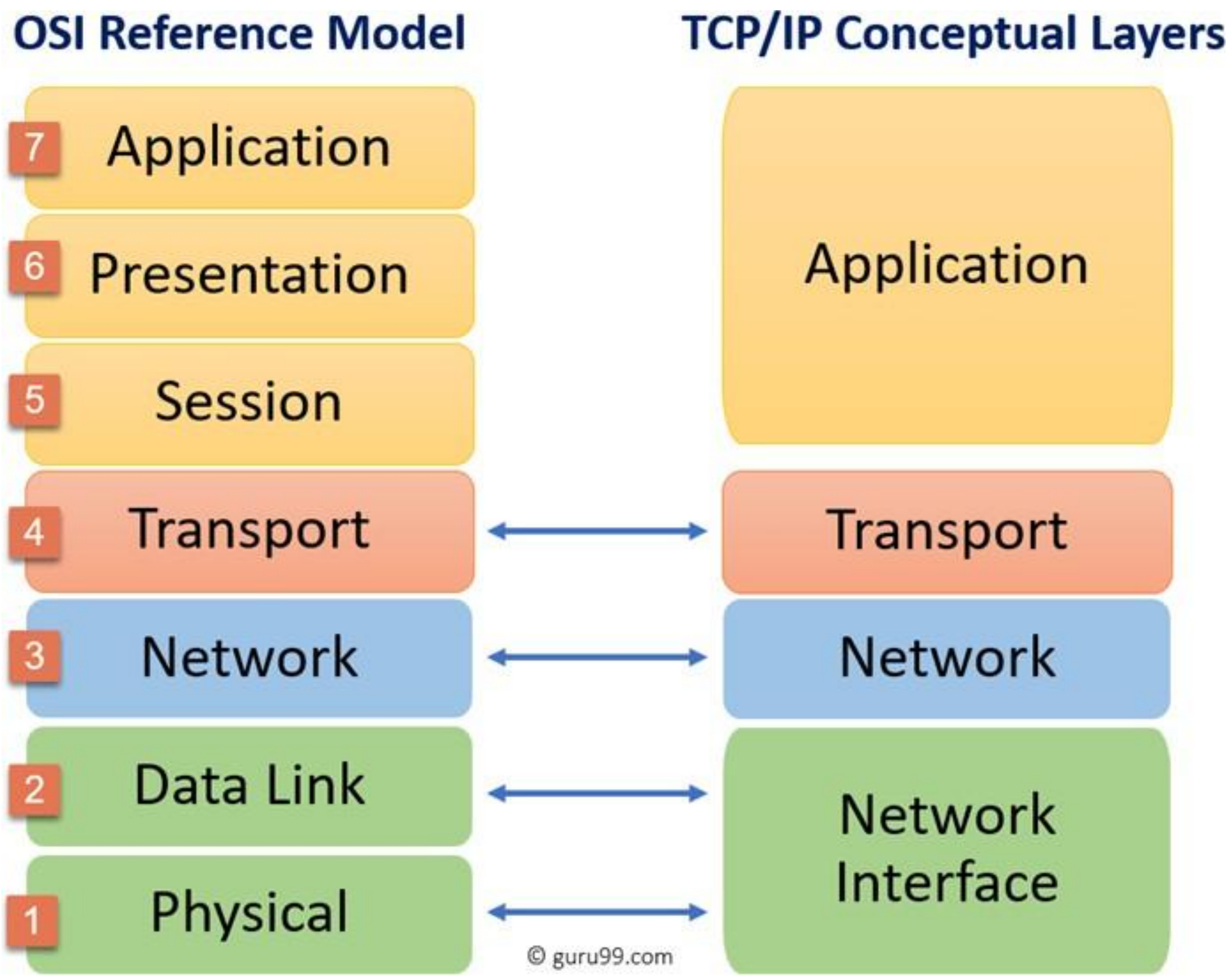


Unicast | Multicast | Broadcast | Unicast | Multicast | Broadcast

What's the difference? Look for it!

Network Reference Models

The Open Systems Interconnection model was introduced by the International Organisation of Standardization in 1984

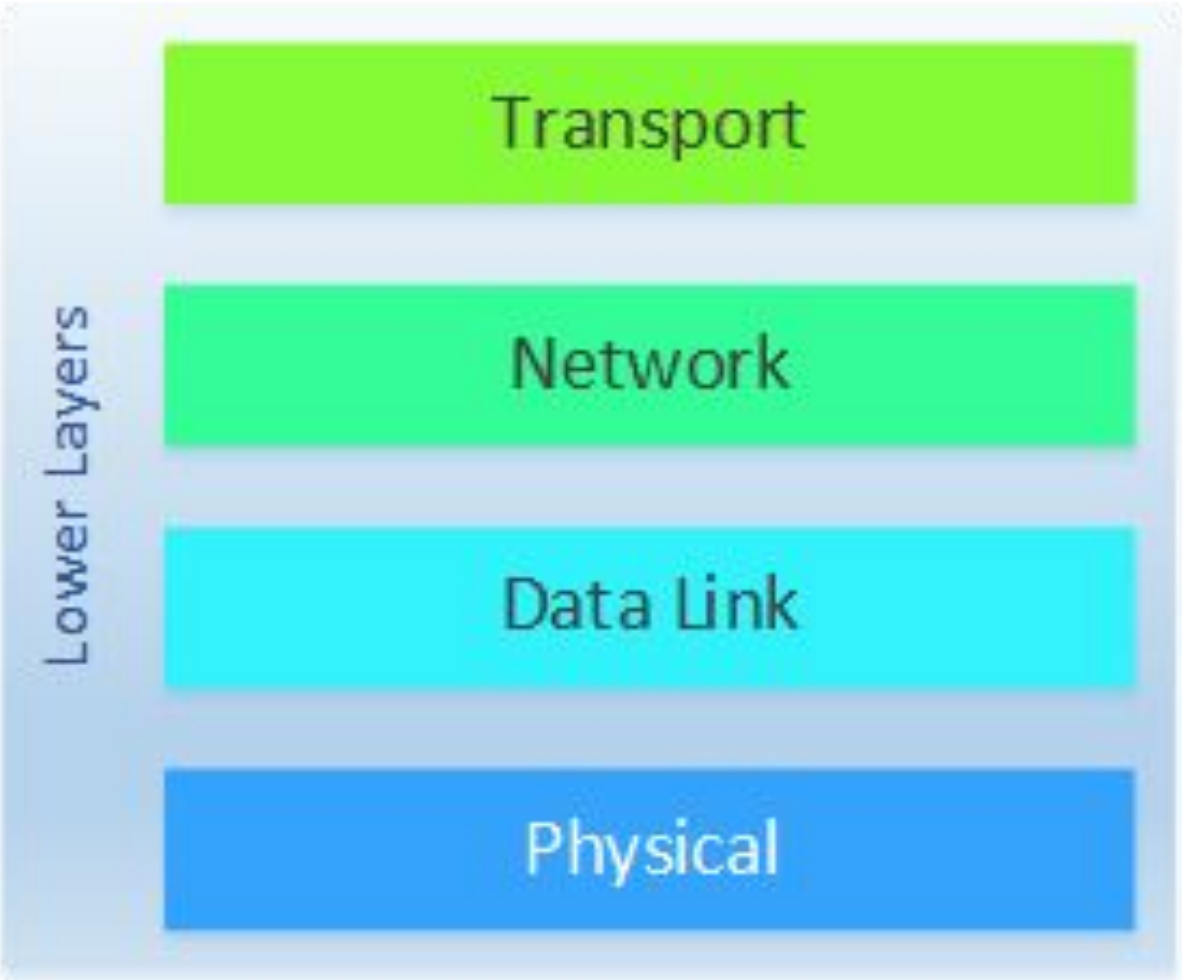


The TCP/IP model was introduced about 10 years before OSI layers

Network Reference Models

Lower Layer

- Responsibility of the network
- Deal with hardware



Upper Layer

- Responsibility of the host
- Deal with software

Application Layer

Application Layer

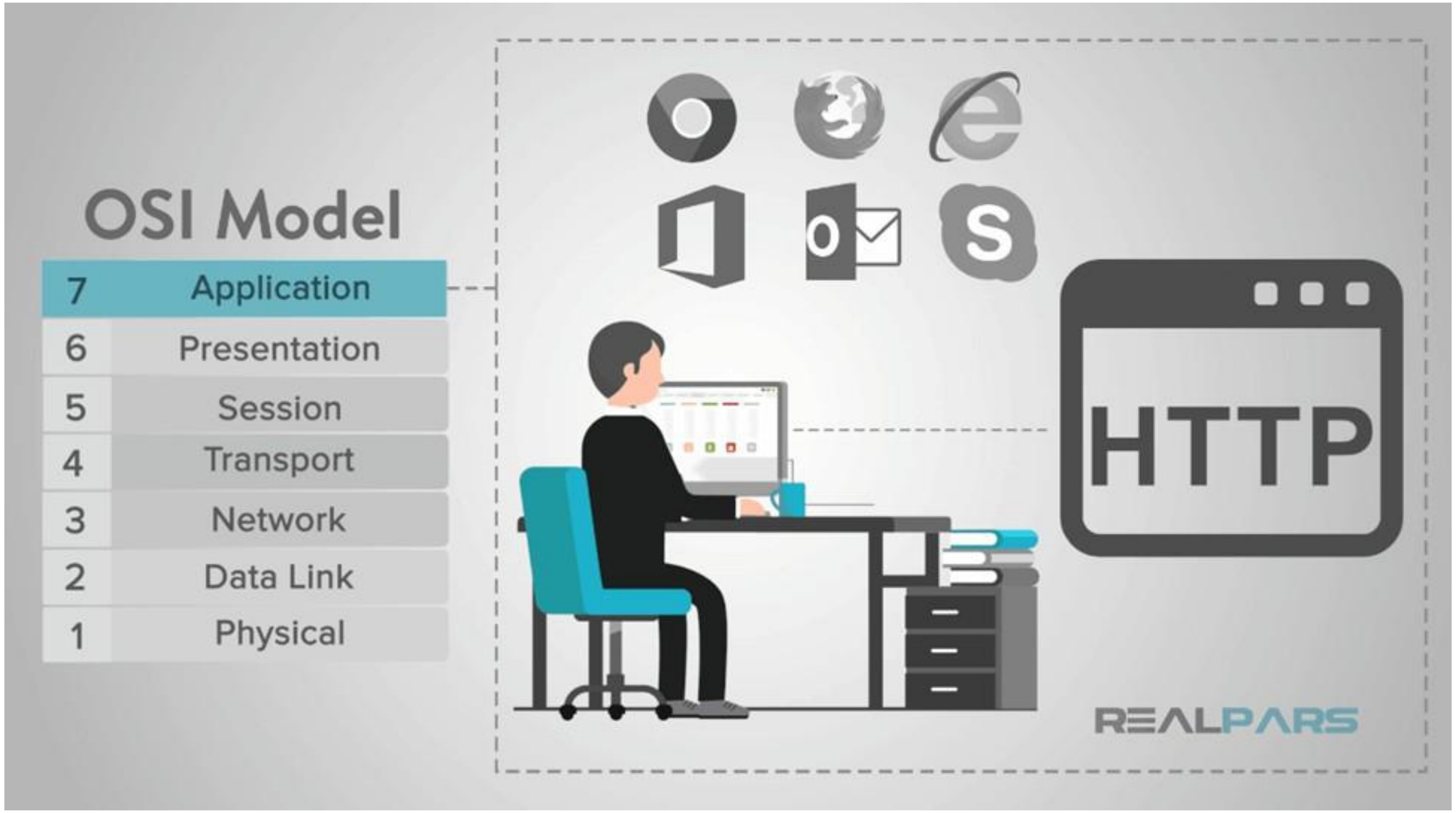
- Interacts directly with software applications
- Closest to end user

DNSPOP3

HTTP/s

TELNET

SMTP FTP

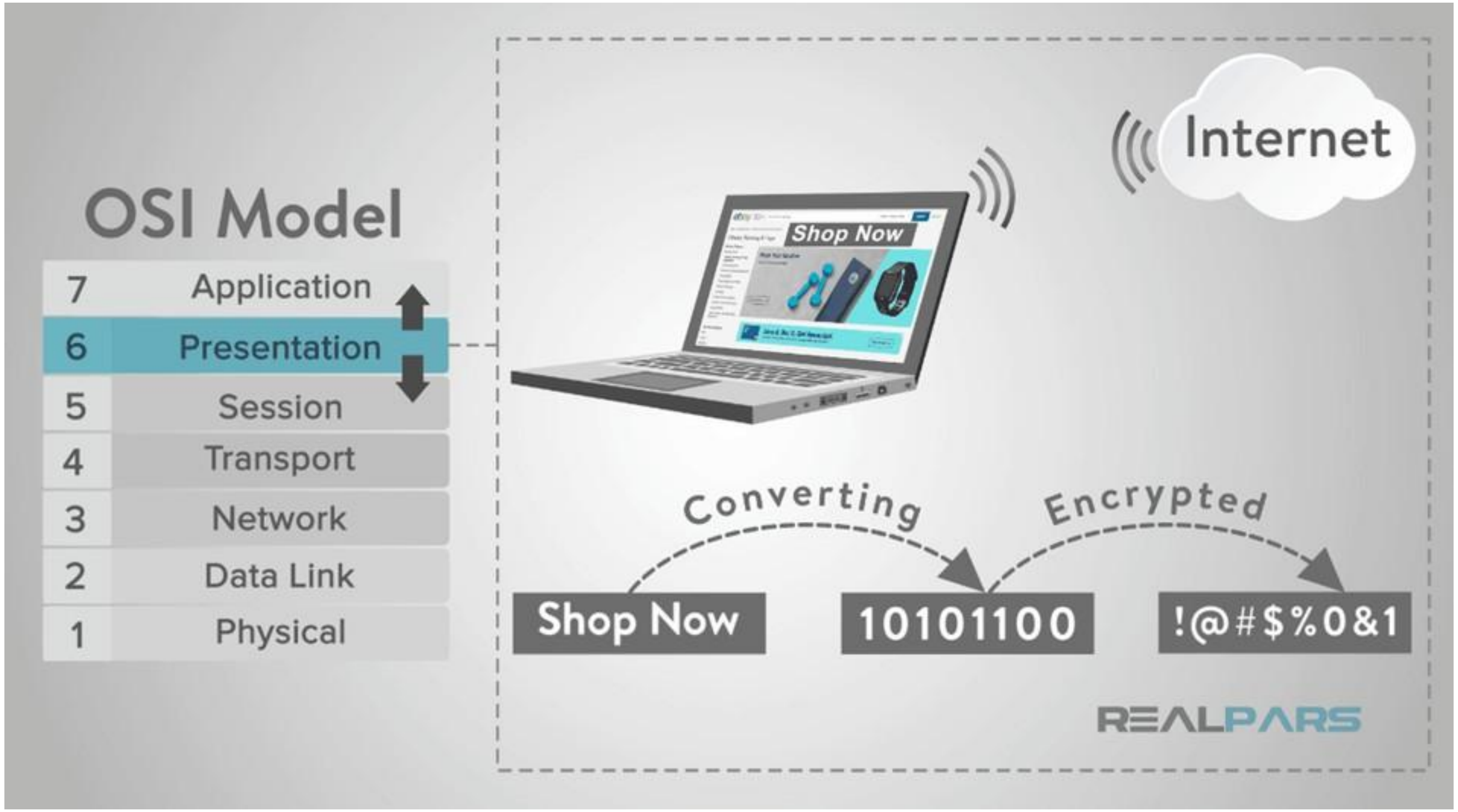


Presentation Layer

Presentation Layer

- Checks the data to ensure it is compatible with the communication
- resources Data
- formatting

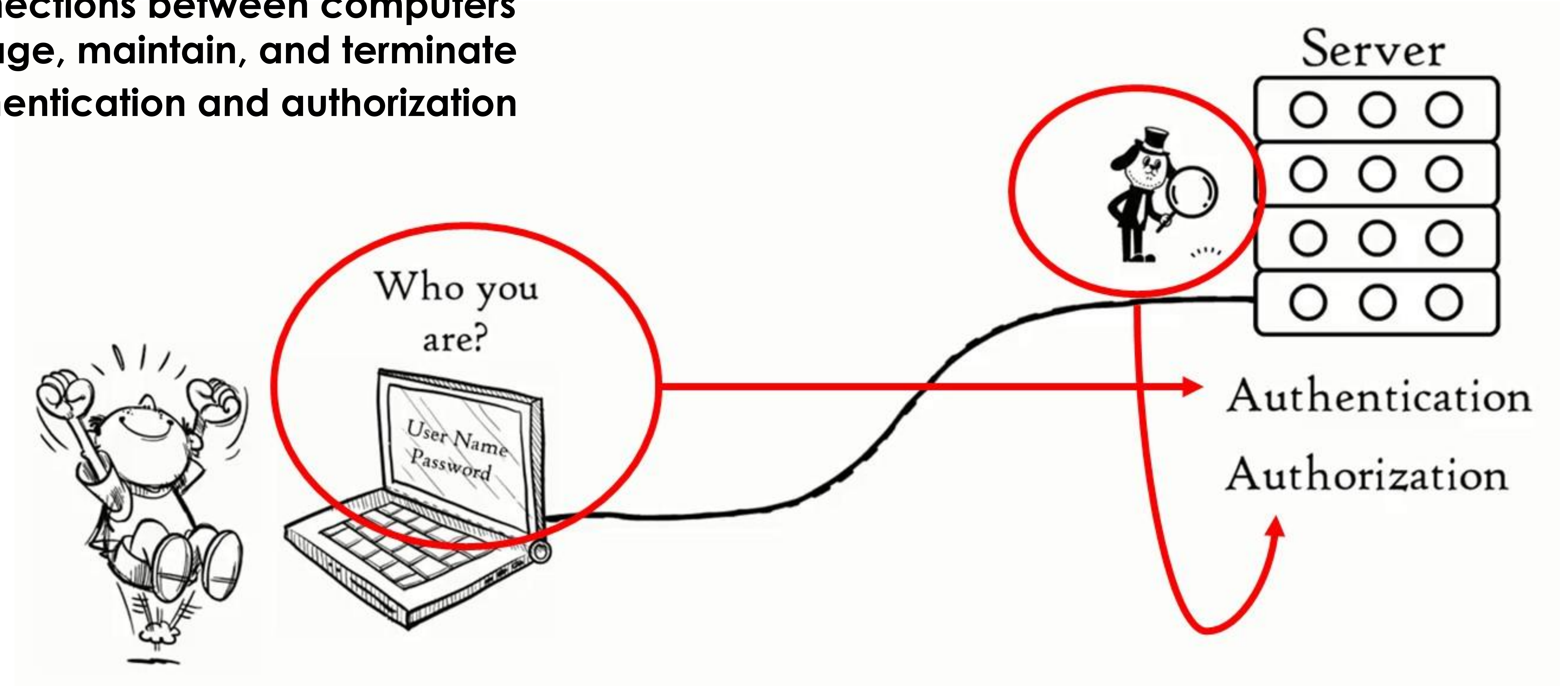
Code conversion



Session Layer

Session Layer

- Controls the connections between computers
- Establishes, manage, maintain, and terminate
- connections Authentication and authorization



API

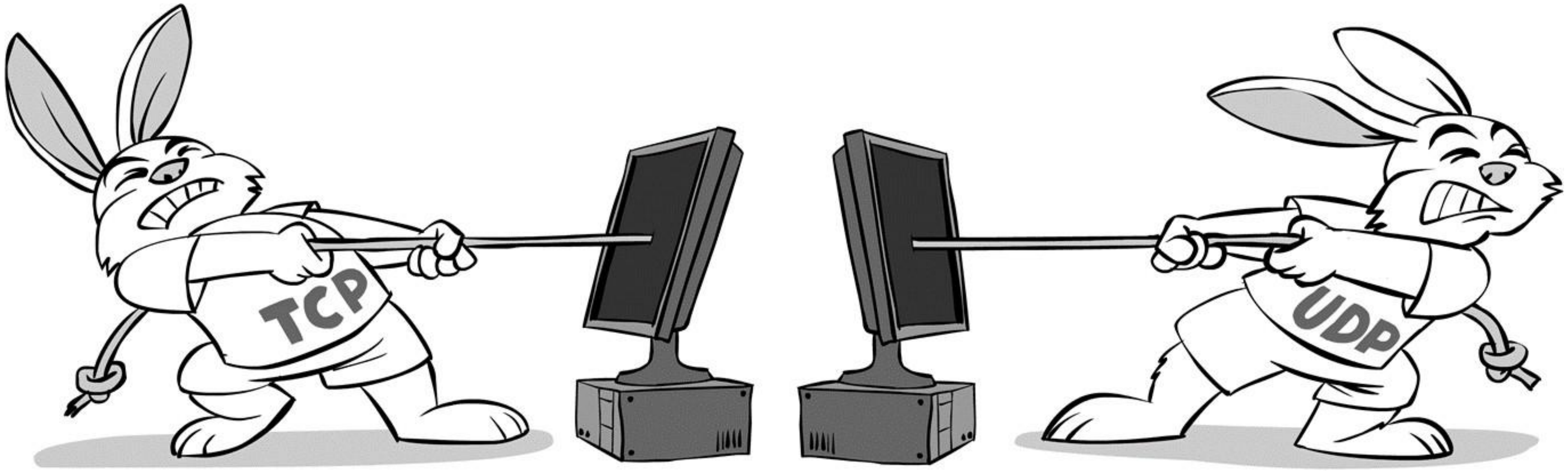
SOCKET

WINSOCK

Transport Layer

Transport Layer

- Provides functions transferring data
- Maintain Quality of Services (QoS) functions
- Ensure the complete delivery of the data

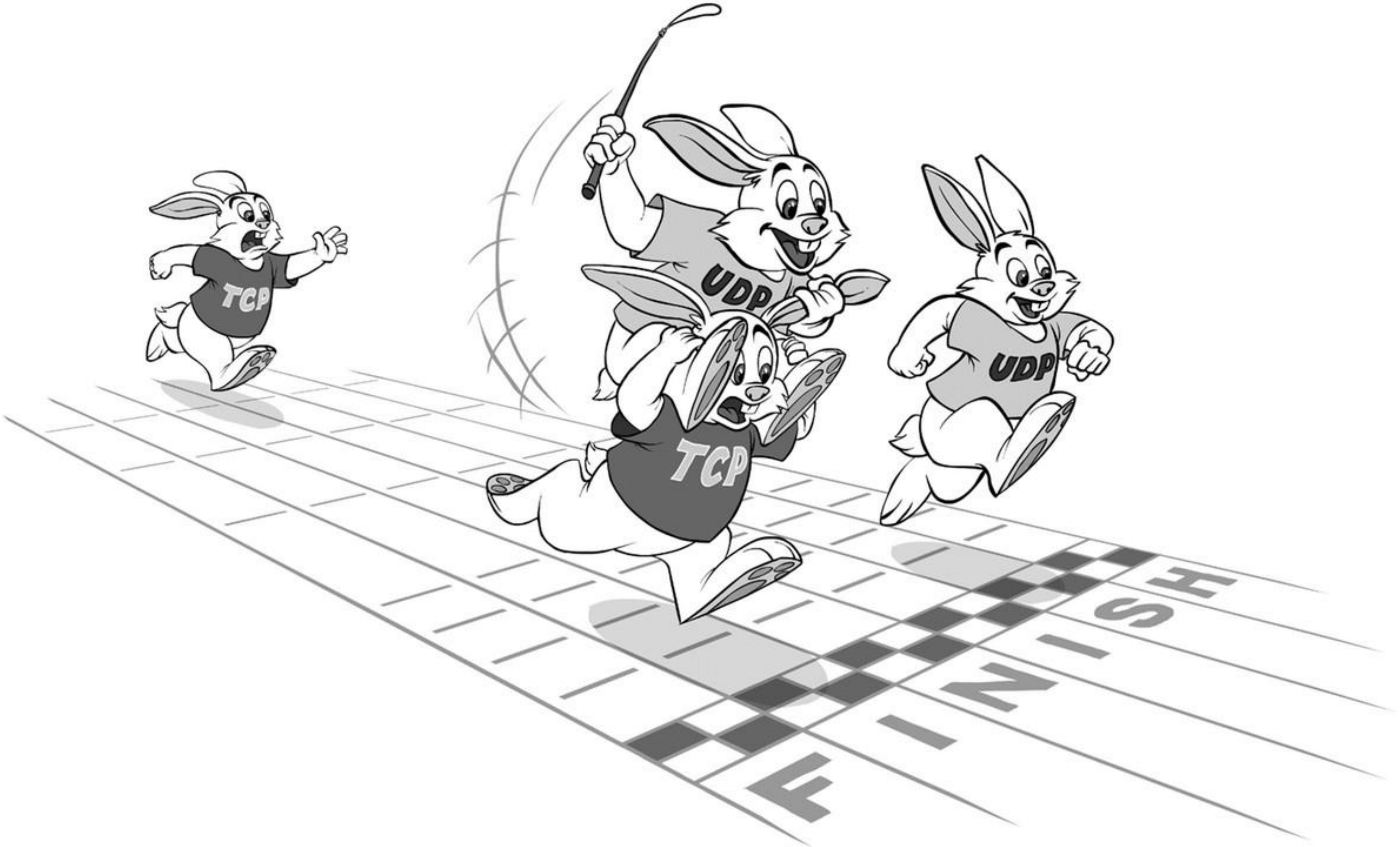


Transport Layer

TCP

vs

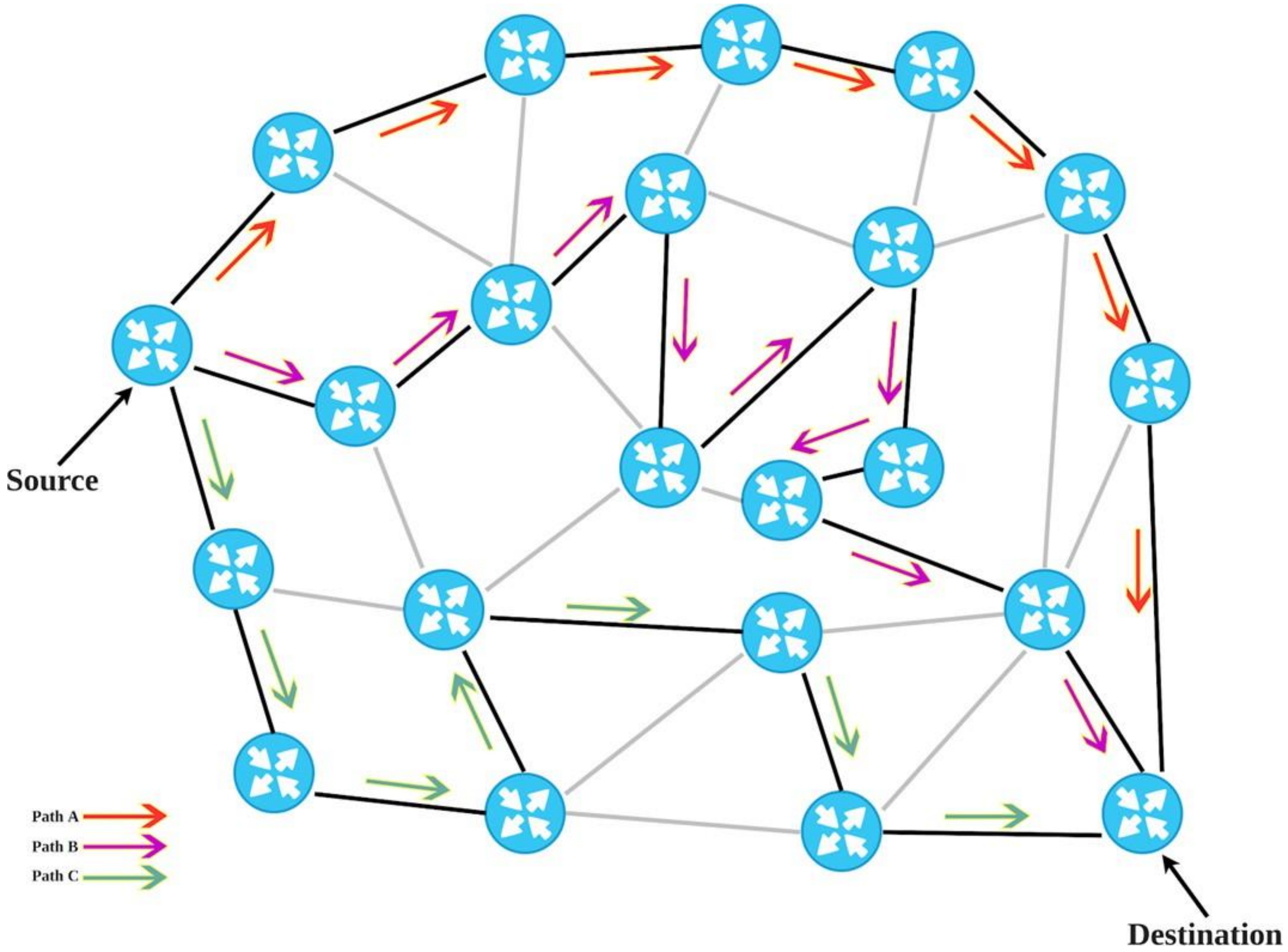
UDP



Network Layer

Network Layer

- Handles packet routing via logical addressing and switching functions
- Find the way to deliver the message to the destination node
- Split the message into several segments



IP

IPSec

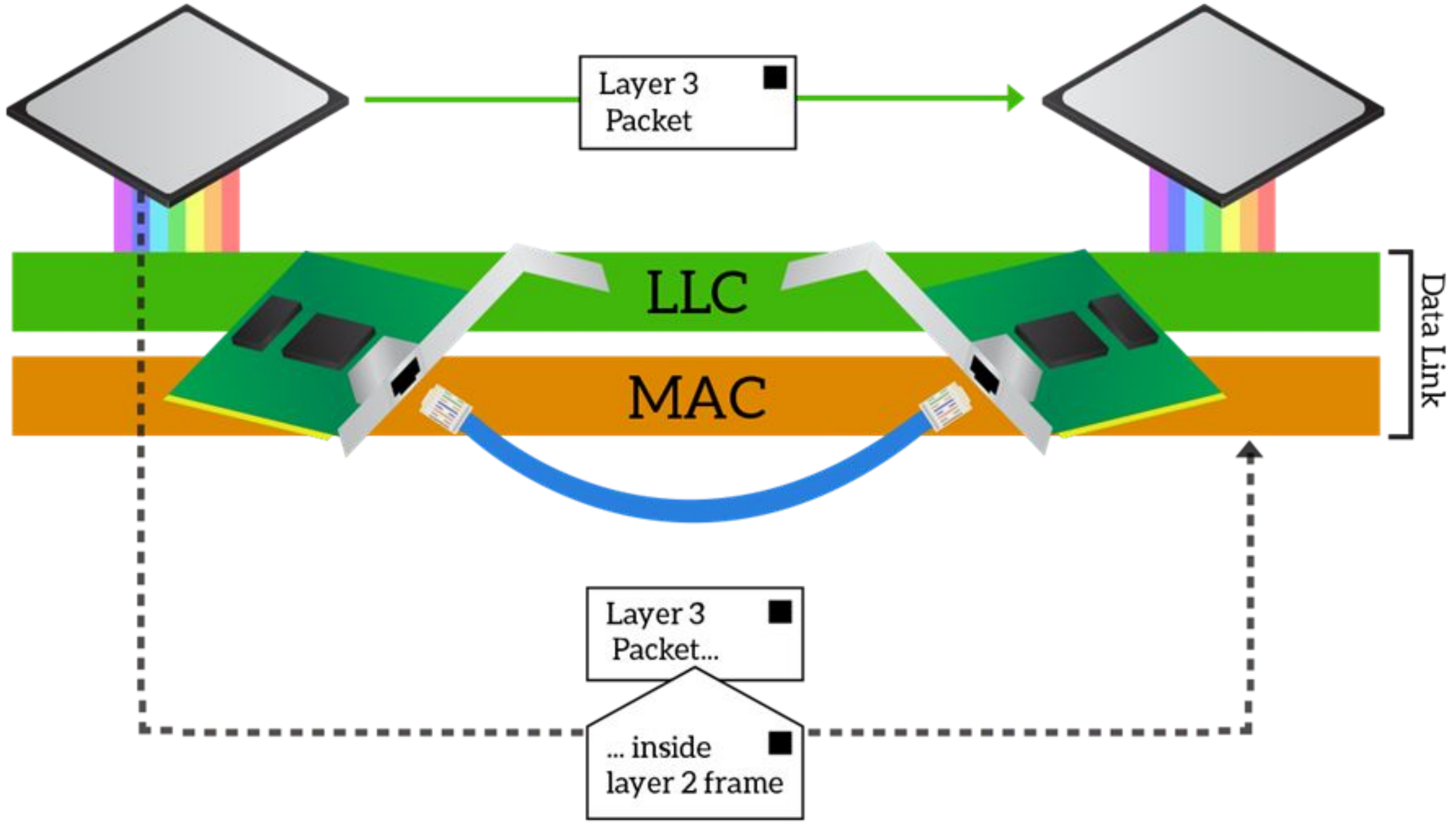
ICMP

IGMP

Data Link Layer

Data Link Layer

- Node-to-node transfer (link between two directly connected nodes)
- Packaging and unpackaging data in frames
- Establish and terminate a connection between 2 physically devices



Ethernet

PPP

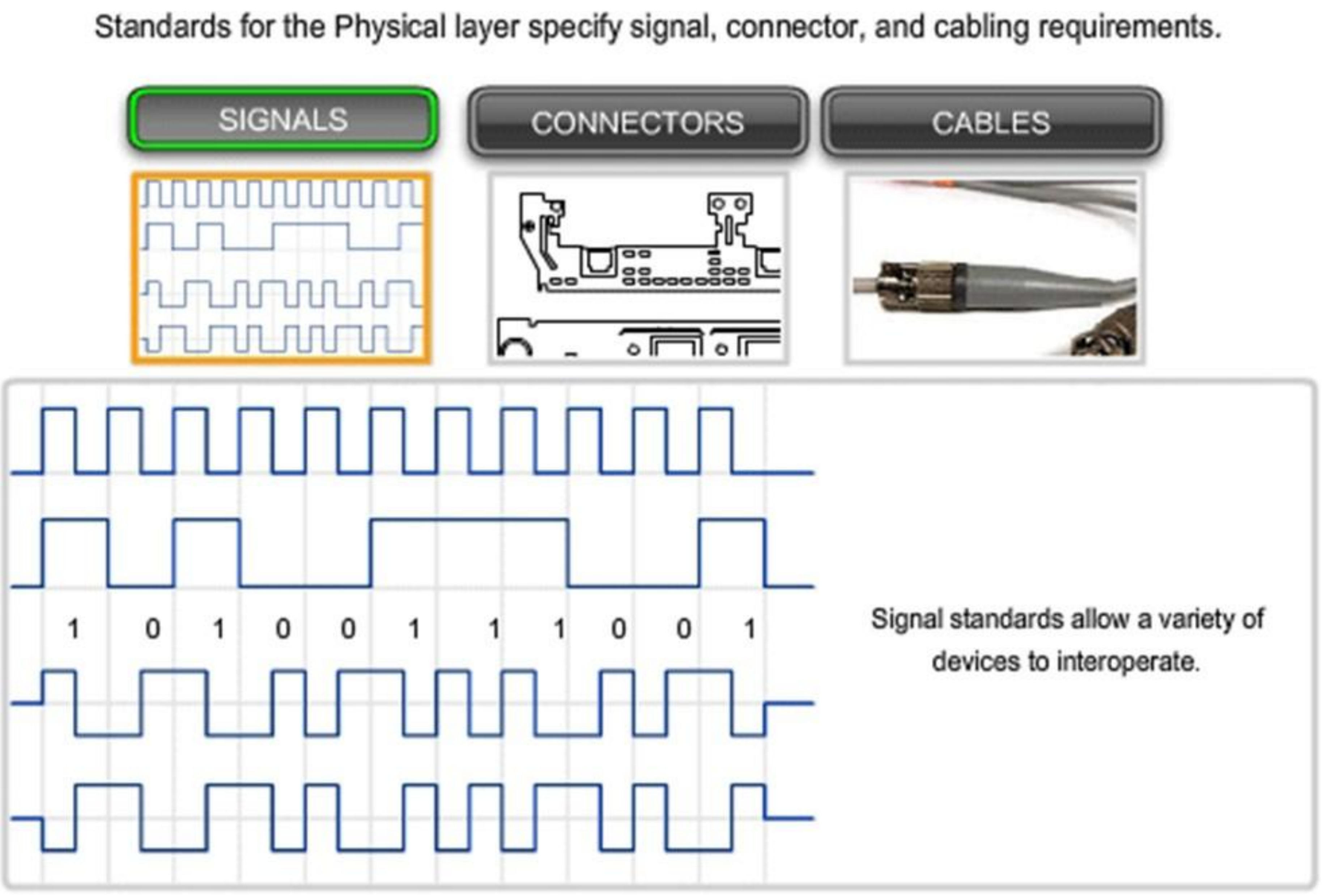
Switch

Bridge

Physical Layer

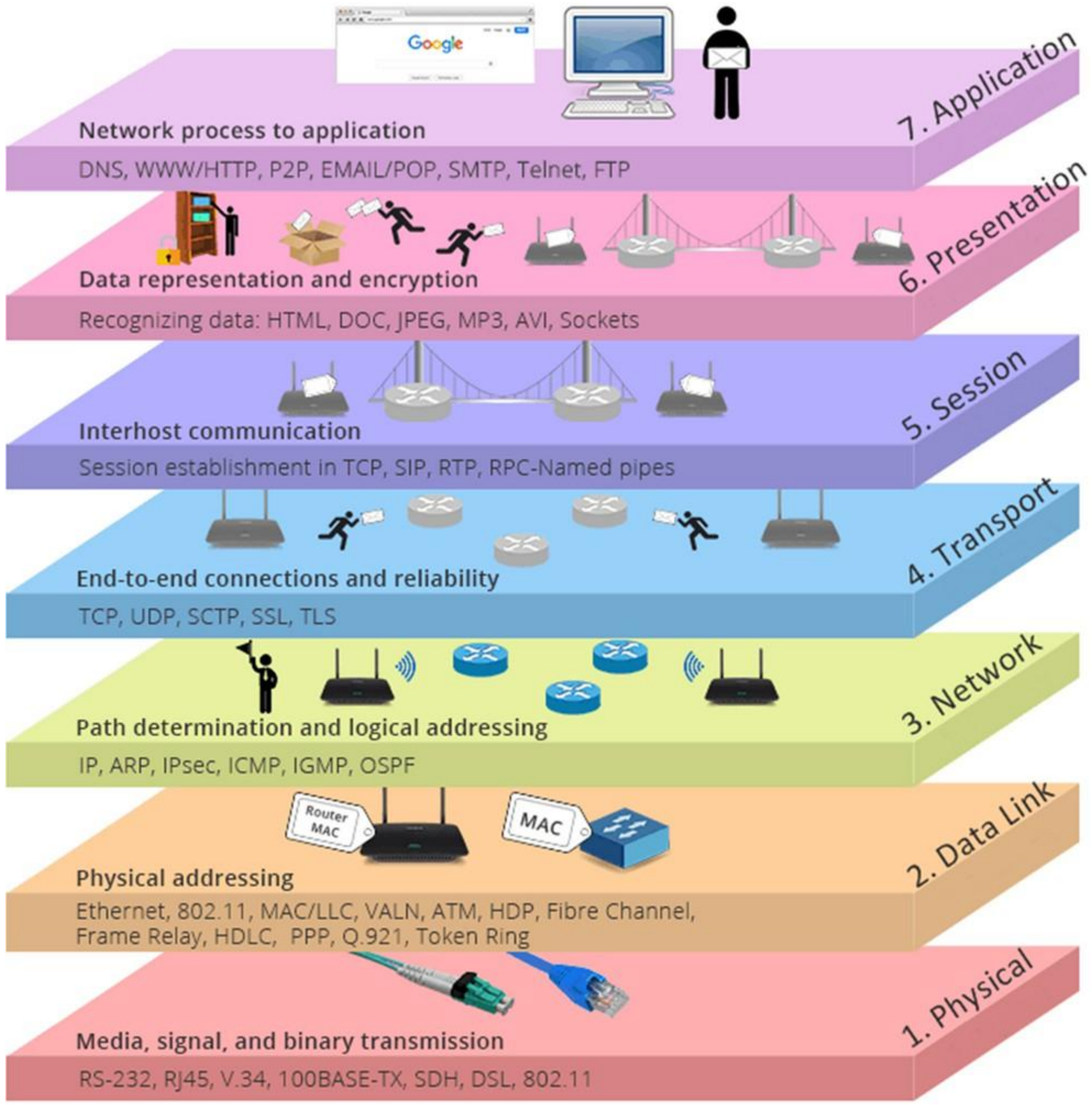
Physical Layer

- Defines the electrical and physical specifications of the data
- connection Transmission and reception of unstructured raw data in a physical medium



OSI Layer Review

1. Physical : media, signal, and binary transmissions
2. Data Link : physical addressing
3. Network : path determination and logical addressing
4. Transport : end-to-end connections & reliability
5. Session : inter host communication
6. Presentation : data representation and encryption
7. Application : network process to application



Tugas

1. Buatlah studi kasus dengan topik “ Network Communication and Protokol dengan pemanfaatan OSI Layer”
2. Gambarkan studi case yang anda angkat dengan sejelas jelasnya (case masalah, infrastuktur yang di bangun, arsitektur, dll)
3. Buatkan / ilustrasikan studi kasus tadi ke (node struktur OSI LAYER)
4. Buatkan ke dalam bentuk PPT

Do you have any questions?

Send it to us! We hope you learned something new.

INSTITUT ASIA